

CHMSU HireMe: Integrated OJT & Career Placement Platform

System Architecture Specification, End-to-End Operational Processes for All User Roles, and Data Flows

Target Platform: CHMSU HireMe System	Backend API: Laravel 12 (Sanctum Auth)	Database: PostgreSQL 16 (27 Core Tables)
Frontend: Decoupled Vite SPAs (Vanilla JS)	Security: Role Middleware & Sanctum Tokens	Document Scope: Full System Process & Flow

1. System Architectural Overview

CHMSU HireMe is an enterprise academic internship (OJT) monitoring and career placement platform engineered for Carlos Hilado Memorial State University. It unifies undergraduate OJT curriculum tracking, Host Training Establishment (HTE) industry partnerships, faculty supervision, and post-graduation career hiring into an automated, geofenced ecosystem.

Architecture Tier	Technology Stack	Functional Responsibilities & Sub-Modules
Client Tier (Frontends)	Vite, Vanilla ES6+ JS, Modular CSS, SVG	Dedicated SPAs: /landing (Public), /login (Auth & Onboarding), /main (Student), /company (HTE), /supervisors (Coordinators), /jobseeker (Alumni), and /admin (Placement & CIER).
Application Tier (REST API)	Laravel 12 API, Sanctum Token Auth, PHP 8.3+	Centralized business controllers: Student, Company, Supervisor, Jobseeker, Admin, OJT, Evaluation, Chat, and Notification. Enforces role security, geofencing, and automated triggers.
Persistence Tier (Database)	PostgreSQL 16 / Eloquent ORM Architecture	27 Core tables partitioned across 7 functional domains: Authentication, Profiles, Verified Portfolio, OJT Lifecycle, Evaluations, Career Jobs, and System Communications.
External Integrations	HTML5 Geolocation API, Cloud Storage, SMTP	Captures latitude/longitude for attendance check-ins; archives bilateral MOA PDF contracts; dispatches real-time appraisal notifications and email alerts.

2. System Actors & Operational Roles

User Actor	Portal Path	Primary Core Responsibilities in the System
Student / Trainee	/main/	Curates verified portfolio; submits pre-deployment clearance drive link; discovers OJT slots; clocks geofenced DTR attendance; tracks required hours (300/486/600 hrs); reviews appraisals; answers exit PEO survey.
Company (HTE)	/company/	Uploads/tracks bilateral MOA with CHMSU CIER; publishes OJT slots & job openings; screens applicants; requests endorsements; schedules interviews; evaluates interns via standardized Likert rubrics.
Supervisor (Coordinator)	/supervisors/	Manages trainee rosters (CSV import); reviews pre-deployment clearance folders; issues formal academic endorsement slips; tracks students on interactive map; audits daily time logs; authors & sends rubrics.
Graduate / Alumni (Jobseeker)	/jobseeker/	Maintains professional career portfolio; updates employment state (Employed/Unemployed/Continuing Studies); applies to industry partner jobs; chats with HR; accepts job offers; participates in tracer surveys.
System Admin (CIER & Placement)	/admin/	System-wide governance; validates company accreditations and active MOA contract files; provisions faculty supervisor accounts; runs skills matching engine; promotes graduates; generates CHED accreditation reports.

3. Detailed System Process for All Users

Each user role undergoes a structured, verifiable lifecycle with explicit validation checkpoints:

3.1. Student / OJT Trainee Process Lifecycle

Step 1: Sign-Up & Academic Onboarding

Initiates registration at /login/. In onboarding wizard, inputs campus (Talisay, Fortune Towne, Alijis, Binalbagan), program (e.g. BSIT), section, batch, and official Student ID Number.

Step 2: Portfolio & Verified Credentials Builder

Under /main/#portfolio, builds digital resume: educational history, technical skills (0-100 rating), work/volunteer experience, verified certificates/awards with attachments, and projects.

Step 3: Pre-Deployment Clearance Submission

Reviews coordinator's clearance checklist (Medical, Insurance, Parent Consent, Barangay Clearance). Submits verified Google Drive folder via PUT /api/student/requirements-drive.

Step 4: OJT Discovery & Expression of Interest

Explores accredited company openings in /main/#ojt. Submits interest via POST /api/ojt/interest/{id}, automatically alerting the supervisor for endorsement review.

Step 5: Supervisor Endorsement & Screening

Supervisor validates standing & MOA, issuing an Endorsement Slip. Company reviews verified portfolio, schedules an interview, and marks student Accepted.

Step 6: Geofenced Daily Attendance Logging (DTR)

Under /main/#ojt-tracker, trainee punches 'Time In' and 'Time Out'. Device GPS is validated against host company geofence coordinates; accumulates hours toward target.

Step 7: Trainee Appraisal & Exit Survey

Upon completing required hours, employer submits student's digital evaluation rubric. Student views appraisal scores and completes exit Program Educational Objectives (PEO) Survey.

Step 8: Transition to Graduate / Jobseeker

Upon completion of degree, the account is promoted to Graduate status, activating the career placement features.

3.2. Host Training Establishment (HTE) / Company Process Lifecycle

Step 1: Self-Registration or Admin Invite

Registers at /login/ or accepts admin invite email. Completes company profile: industry sector, company size, physical headquarters coordinates, and HR coordinator.

Step 2: MOA Partnership Verification

Submits Memorandum of Agreement (MOA) partnership request (POST /api/company/request-moa) or coordinates with CHMSU CIER to upload bilateral notarized MOA documents.

Step 3: Vacancy Management (OJT & Career Jobs)

Publishes OJT postings (slots, course requirements, allowances, work setup) and full-time career job openings for graduating students and alumni jobseekers.

Step 4: Candidate Screening & Endorsement Check

Reviews student portfolios; confirms supervisor endorsement; schedules screening interviews (Google Meet or On-Site) via POST .../schedule-interview/{interestId}.

Step 5: Trainee Deployment & Instructions

Accepts trainee and provides onboarding guidelines: start date, reporting department, and designated mentor (POST .../set-ojt-start/{interestId}).

Step 6: Digital Evaluation Rubric Submission

Completes standardized 5-point Likert appraisal form dispatched by supervisor (POST /api/company/evaluations/{id}/submit) with narrative recommendations.

Step 7: Career Recruitment Pipeline

Processes alumni job applicants across hiring stages (Reviewing, Shortlisted, Interview, Offered, Hired), extending official digital job offers.

3.3. Faculty Supervisor / Academic Coordinator Process Lifecycle

Step 1: Portal Access & Department Scope

Supervisor logs in at /supervisors/. Portal auto-filters students, companies, and postings matching their supervised academic program (e.g. BSIT).

Step 2: Trainee Roster Management & Batch CSV Import

Manages enrolled student lists. Can bulk-import class rosters via CSV (POST /api/supervisor/students/import), automatically provisioning student accounts.

Step 3: Clearance Checklist Review & Approval

Assigns pre-deployment requirements. Audits student Google Drive submissions and marks documents Approved, Pending, or Returned for Revision.

Step 4: Academic Endorsement Issuance

Reviews student company applications. Validates company MOA status, then generates and issues official university Endorsement Slips (POST /api/supervisor/recommend/{id}).

Step 5: Geofenced Attendance Monitoring & Live Map

In /supervisors/#monitoring-map, monitors active trainees across company locations. Inspects punch times, GPS coordinates, and flags abnormal logs.

Step 6: Evaluation Rubric Authoring & Dispatch

Builds standardized 5-point Likert appraisal rubrics (evaluation_templates). Dispatches evaluation links individually or in batch to company mentors with reminder alerts.

Step 7: OJT Grade Calculation & Academic Clearance

Reviews completed evaluations, verifies total rendered hours against academic requirements, and submits completed grades to university registrar.

3.4. Graduate / Alumni / Jobseeker Process Lifecycle

Step 1: Alumni Profile & Employment Status

Logs into /jobseeker/. Sets graduation year, degree program, and current employment state (Employed, Unemployed, Self-Employed, Continuing Studies).

Step 2: Career Resume & Portfolio Curation

Maintains career resume with verified credentials, GitHub/portfolio links, certifications, and downloadable PDF resume.

Step 3: Job Discovery & Skills Matching

Searches verified employer vacancies and external career postings. Algorithmic matching displays percentage fit with candidate competencies.

Step 4: One-Click Application & Real-Time Chat

Applies to vacancies with one click (POST /api/jobseeker/apply/{id}). Engages in direct message chat with hiring HR officers.

Step 5: Interview Participation & Offer Decision

Monitors interview appointments, attends selection rounds, and officially Accepts or Rejects offers (POST /api/jobseeker/applications/{id}/accept-offer).

Step 6: Longitudinal Alumni Tracer Surveys

Ongoing employment status updates continuously feed into institutional tracer reports for CHED and SUC Leveling.

3.5. System Administrator (Placement & CIER) Process Lifecycle

Step 1: Institutional Governance & KPI Dashboard

Monitors university-wide placement statistics at /admin/: total active OJT placements, MOA compliance rates, active job listings, and alumni employment percentages.

Step 2: HTE Partner Accreditation & MOA Contract Tracking

Evaluates company registration applications, uploads notarized MOA PDF contracts, monitors expiration dates, and issues renewal notifications.

Step 3: Academic Master Data & Supervisor Provisioning

Creates faculty supervisor accounts, assigns academic programs and departments, and executes bulk CSV student/alumni imports.

Step 4: Algorithmic Skills Matching Engine

Runs the skills matching engine (GET /api/admin/skills-matching) comparing student skill vectors with employer job listings to identify regional curriculum gaps.

Step 5: Annual Graduate Batch Promotion

Executes automated graduation promotion (POST /api/admin/students/promote-graduates) to convert graduating students to alumni jobseekers.

Step 6: Institutional Accreditation & Broadcasts

Generates CHED accreditation reports and PEO compliance summaries; broadcasts campus-wide alerts to students and faculty.

4. End-to-End System Flows & Operational Interactions

The platform coordinates five core multi-party operational workflows across actors:

4.1. Core Flow 1: OJT Internship Deployment & Lifecycle Flow

Sequential lifecycle from company accreditation to clearance, endorsement, screening, and deployment:

Seq	Source	Target	Operation & API Endpoint	Data State Transformation
1	Company	Admin (CIER)	Submits company profile & requests MOA partnership. POST /api/company/request-moa	moa_status = 'pending'
2	Admin (CIER)	Company	Approves accreditation & uploads bilateral MOA PDF. POST /api/admin/companies/{id}/moa	moa_status = 'active'
3	Company	Student	Publishes OJT slot posting with target skills. POST /api/company/ojt-postings	New ojt_postings record created.
4	Supervisor	Student	Assigns pre-deployment clearance checklist. POST /api/supervisor/requirements/assign	student_ojt_requirements initialized.
5	Student	Supervisor	Submits cloud folder link for clearance verification. PUT /api/student/requirements-drive	Clearance status set to 'submitted'.
6	Supervisor	Student	Reviews documents and clears student for deployment. PUT /api/supervisor/requirements/{id}/review	Clearance status set to 'approved'.
7	Student	Company	Expresses interest in accredited OJT slot. POST /api/ojt/interest/{id}	student_ojt_interests state = 'interested'.
8	Company	Supervisor	Reviews student portfolio; requests official endorsement. POST ../request-endorsement/{interestId}	Interest status = 'endorsement_requested'.
9	Supervisor	Company	Issues formal academic endorsement slip. POST /api/supervisor/recommend/{id}	Interest status = 'endorsed'.
10	Company	Student	Schedules interview & marks student accepted. POST ../accept-after-interview/{interestId}	Interest status = 'accepted'.
11	Company	System	Establishes official internship commencement. POST ../set-ojt-start/{interestId}	Active ojt_records record created.

4.2. Core Flow 2: Daily Time Record (DTR) & Geofenced Attendance Flow

Step	Actor Action	System Process & Geofence Verification	Resultant State
1	Student: Clock In	Student taps 'Time In' on mobile/web portal. HTML5 Geolocation captures latitude/longitude. Calls POST /api/student/ojt-tracker/log-in.	System verifies GPS proximity to host company geofence coordinates.
2	API: Punch Record	Server validates no open log exists today. Records in_time, in_latitude, in_longitude into time_logs table.	New time_logs record created in active shift state.
3	Student: Clock Out	At shift end, student taps 'Time Out'. Captures exit coordinates. Calls POST /api/student/ojt-tracker/log-out.	Records out_time; computes total decimal hours rendered.
4	API: Hours Sync	Calculates shift duration minus standard break. Increments rendered_hours in parent ojt_records table.	Progress bar increments toward curriculum target (e.g. 486 hrs).
5	Supervisor: Audit	Supervisor inspects live monitoring map and audits daily time logs at /supervisors/#monitoring-map.	Supervisor verifies authentic attendance or flags anomalies.

4.3. Core Flow 3: Trainee Performance Evaluation & Appraisal Flow

Phase	Acting Entity	Procedure Details	Artifact / Output
1. Rubric Authoring	Faculty Supervisor	Creates university evaluation templates (evaluation_templates) with criteria across categories (Technical Competence, Punctuality, Attitude, Teamwork) using 5-point Likert scales.	Standardized institutional evaluation template.
2. Dispatch Appraisal	Faculty Supervisor	Dispatches evaluation requests to company mentors (POST /api/supervisor/evaluations/send/{student_id} or batch). Triggers automated notification to employer HR.	student_evaluations record initialized (status = 'pending').
3. Appraisal Submission	Company Mentor	Mentor opens digital form (/company/#evaluations), assigns 1–5 ratings to criteria, and inputs narrative feedback and strengths/recommendations.	student_evaluation_answers populated; evaluation marked 'completed'.
4. Grade Computation	Faculty Supervisor	Coordinator reviews aggregated overall rating score (e.g. 4.85/5.00), factors in attendance percentage, and generates final academic grade.	Final numerical OJT grade recorded in academic ledger.
5. Feedback & PEO Survey	Student Trainee	Trainee views completed appraisal report and completes the institutional Program Educational Objectives (PEO) Survey.	Closed OJT record; student cleared for graduation.

4.4. Core Flow 4: Graduate Career Recruitment & Hiring Pipeline

Stage	Actor Action	Platform Mechanism & API Endpoint	Pipeline State
1. Vacancy Posting	Company	Publishes job listing specifying required competencies, salary range, and job level. POST /api/company/jobs	Active job_listings opening published.
2. Candidate Application	Jobseeker / Alumni	Applies using verified digital profile and portfolio link. POST /api/jobseeker/apply/{jobId}	job_applications status = 'applied'.
3. Screening & Chat	Company & Candidate	HR reviews application and begins direct message thread. POST /api/chat/{key}/messages	job_applications status = 'reviewing'.
4. Interview Scheduling	Company	Schedules interview with date, time, and meeting link. POST /api/company/applications/{id}/interview	New interviews appointment record created.
5. Offer & Placement	Company & Candidate	Company issues official offer. Candidate accepts or rejects. POST /api/jobseeker/applications/{id}/accept-offer	Status updated to 'hired'; alumni employment record updated.

4.5. Core Flow 5: Institutional Accreditation, MOA & Alumni Tracer Flow

Accreditation Area	System Function & Execution	Regulatory / Academic Output
MOA Legality & Safety	CIER Admin validates company legal standing, uploads notarized MOA PDF contracts, and tracks validity expiration dates to ensure all active trainee placements comply with CHED Memorandum Order No. 104.	CHED MOA Compliance Audit Ledger & Active Industry Partners Directory.
Skills Gap Analytics	Admin skills matching algorithms analyze student competency vectors against industry job requirements to detect curriculum deficiencies.	Curriculum Enhancement Recommendations for College Academic Council.
Longitudinal Tracer	Platform tracks employment trajectories of graduates across graduation batches, capturing employment status, industry alignment, and job acquisition timelines.	Annual SUC Leveling, CHED Institutional Sustainability Assessment (ISA) Tracer Report.
PEO Survey Analysis	Aggregates exit and alumni surveys assessing Program Educational Objectives and Student Outcomes.	Accreditation Commission (AACCUP) Compliance Exhibits.

5. Database Architecture & Entity Relational Mapping

The relational database schema is structured into 27 core application tables across seven operational domains:

Domain	Entity Table Name	Architectural Purpose, Foreign Keys & Business Rules
Authentication (auth)	users personal_access_tokens	Central identity table containing user login credentials, Bcrypt-hashed password, primary role flag (student, company, supervisor, graduate, jobseeker, admin), and onboarding status.
Actor Profiles (profile)	student_profiles company_profiles supervisor_profiles graduate_profiles	Role-specific metadata tables linked 1:1 to users.id. Houses university student IDs, campus branches, degrees, company MOA validity states/dates, department allocations, and alumni employment statuses.
Portfolio (portfolio)	student_education student_skills student_experiences student_achievements portfolio_projects	Verified digital resume entities: academic history, competency skill ratings (0-100), prior work/internship experiences, certificates/awards with attachments, and showcase portfolio projects.
OJT Lifecycle (ojt)	ojt_postings student_ojt_interests student_ojt_requirements ojt_records time_logs	Internship management: openings posted by HTEs, application interest records, pre-deployment clearance checklists, active training contracts, and geofenced daily attendance punch logs.
Evaluation (eval)	evaluation_templates evaluation_questions student_evaluations student_evaluation_answers	Performance evaluation rubrics: standardized templates created by faculty, question criteria, employer appraisal instances, and itemized 5-point Likert question scores.
Career Jobs (job)	job_listings job_applications interviews	Graduate career recruitment: company job openings, alumni applications, interview appointments, and formal job offers.
Communication (comm)	chat_messages app_notifications	Direct bidirectional chat threads between candidates and employers; automated event notifications and campus-wide broadcast announcements.

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